# Precept 6: Information Theory

November 10, 2026

## Problem

1. The table below gives the joint probability distribution for random variables $X$ and $Y$

| $P(X,Y)$ | $Y=0$ | $Y=1$ |
|---|---|---|
| $X=0$ | $1/2$ | $1/4$ |
| $X=1$ | $0$ | $1/4$ |

   (a) Compute the marginal distributions $P(X)$ and $P(Y)$.

   (b) Compute the conditional probability table $P(X \mid Y)$.

   (c) Compute $H(X \mid Y)$ using

$$H(X \mid Y) = -\sum_{x,y} P(x,y) \log_2 P(x \mid y)$$

   (d) Now compute the conditional probability table $P(Y \mid X)$ and use it to find $H(Y \mid X)$.

   (e) Is $H(X \mid Y) = H(Y \mid X)$?

2. You have 12 identical-looking coins. One is fake, and it's either ever so slightly heavier or lighter than the rest. You don't know which.

   (a) How many possible "states of the world" are there?

   (b) How many questions should you expect to ask to figure out which coin is heaviest?

3. Remember the chain rule for probabilities from last time?

$$P(x,y) = P(x \mid y)P(y)$$

   Pretty cool right? There is also a chain rule for entropy

$$H(X,Y) = H(X|Y) + H(Y)$$

   (a) Prove it. (*Hint:* The definition of entropy is $H(X) = -\mathbb{E}\left[\log P(X)\right]$)

   (b) Use the chain rule for entropy to show mutual information is symmetric

$$I(X;Y) = I(Y;X)$$

# Solution

### Question 1

### Question 2

### Question 3